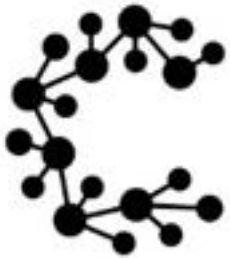


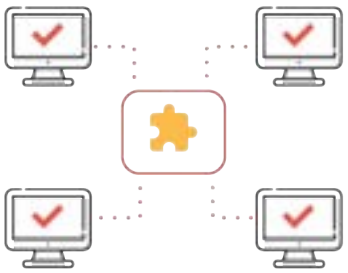
CASPERLABS

A fully decentralized, truly scalable blockchain

Powered by the first implementation of
CBC Casper proof-of-stake protocol.

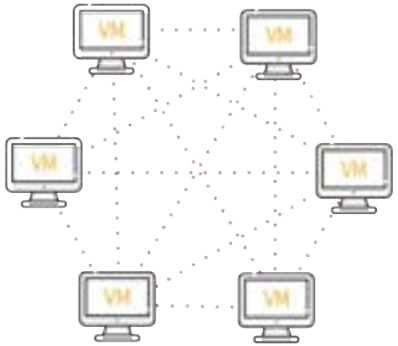
First Gen Blockchains Solved Some Fundamental Issues

The Bitcoin Blockchain



Decentralized trust is built by each node or validator referencing a copy of a ledger - a great solution for maintaining balances.

The VM-Based Blockchain



Pioneered by Ethereum, referencing the state of a virtual machine (VM), made nodes “smart” and capable of processing contracts.

The Ethereum VM was a serialized
“proof-of-work” system
which has **limited the network to ~15 TPS.**

Scale has held blockchain back.

Today's platforms can't scale in technology, usability, value, or trust.

This has forced compromises in performance, security, and decentralization.

Creating complexity at the command line.
Risking critical data to single points of failure.

Problems with Current VM-Based Blockchains

PERFORMANCE
Faster platforms sacrifice decentralization and security to achieve speed.

SECURITY
Smart contracts are complex to build and difficult to verify, resulting in unexpected outcomes.

PRICING
Unpredictable prices make it impossible for businesses to plan, while uniform pricing limits flexibility.

TOOLS
Proprietary languages, tools, and approaches have a steep learning curve, slowing adoption velocity.

CASPERLABS

Blockchain for the builders

The blockchain platform purpose-built
to scale opportunity for everyone, forever.

Removing the Barriers that Prevent Adoption

SCALE WITHOUT SACRIFICE

—

Providing fully decentralized, truly scalable blockchain services made for real-world speed and security

FRIENDLY TO USE

—

Familiar tools, open architectures, and expert support to make any developer a blockchain developer

OPEN TO THE WORLD

—

A public and permissionless software platform developed with radical transparency

FUTURE-PROOF FOUNDATION

—

Establishing upgradeable infrastructure to support enduring and equitable innovation

THE TECH FOR TODAY

THE VISION OF TOMORROW

Journey to Mainstream Adoption

1st Generation Blockchains	CasperLabs
Slow and inefficient *OR* centralized	**Fast and efficient *AND* decentralized**
Unpredictable pricing for apps	**Predictable economics for business**
Steep learning curve and niche tools	**Leverage popular languages, tools, and workflows**
Limited upgradeability	**Future-proof foundation**

What We Are Building

- **Innovative protocol and virtual machine** designed to support:
  - Concurrent execution
  - Sidechains
  - Sharding

- CBC Casper proof-of-stake to validate transactions quickly while **maintaining decentralization that scales for real-world business.**

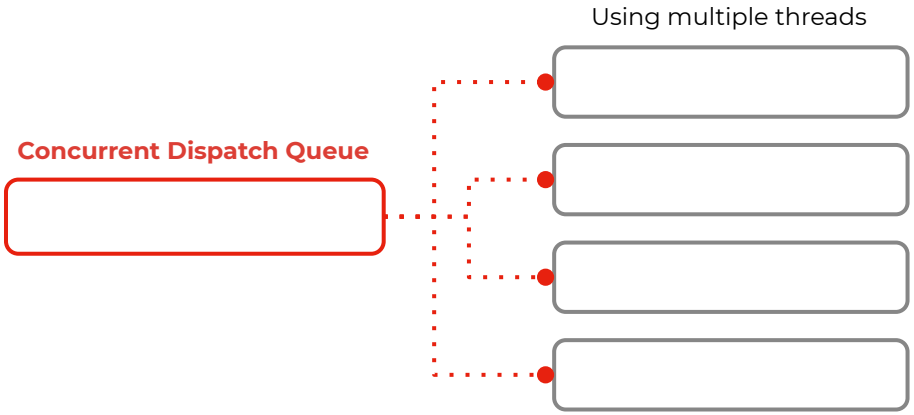
- **Heterogeneous sidechains** enable the creation of custom chains to serve specific market needs such as speed, geographic location, security, and more.

- Enabling developers to use the skills they already have, so **every developer is a blockchain developer.**

Innovation That Scales

CONCURRENT ORCHESTRATION

Our VM achieves performance via mathematical and computational fundamentals



Concurrent orchestration allows greatly increased throughput. Instead of tasks waiting on other tasks, they can execute at the same time. This is done by using an execution engine that supports concurrent processing.

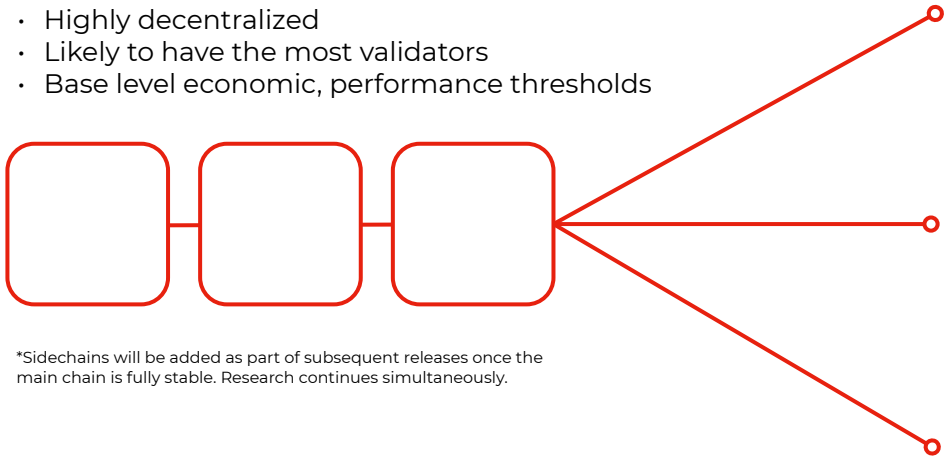
Innovation That Scales

SIDECHAINS

Sidechains* allows multiple configurations with scaling as a bonus

Root Shard

- Highly decentralized
- Likely to have the most validators
- Base level economic, performance thresholds



*Sidechains will be added as part of subsequent releases once the main chain is fully stable. Research continues simultaneously.

Storage Shard

- Globally distributed
- Hardware focused on storage
- Slow and cheap processing

Asia dApp Shard

- Outlay Asia concentrated
- Hardware focused on processing
- Fast low latency within region

Securities Shard

- Outlay co-located close to exchanges
- Compliant with domicile requirements
- Very fast, low latency

Without Sacrificing Decentralization or Security

Open to the World

A public and permissionless software platform developed with radical transparency.

Participation in Consensus

Low barriers to entry and a rotating consensus mechanism not controlled by committees or voting, promote decentralization and enhance security.

Correct by Construction

Algorithms written with fundamental, verifiable mathematical properties enable higher security through provable, predictable behaviors.

Proof-of-Stake for Large-Scale Public, Decentralized Networks

Key Features of CBC Casper:

Proposed by **Vlad Zamfir** of the Ethereum Foundation to address scalability and energy consumption issues.

Punishes Key Bad Behavior

- Cartelization / censorship
- Equivocation
- Reversion
- Invalidity

Abstract and Future Proof

- Ready-built for sharding
- Built with upgradability in-mind
- Flexible payment model

Increases Decentralization

- Can have Min/Max staking amounts
- Can raise participation requirements with minimal overhead increase
- Allows people with limited hardware (within reason) to participate in securing the network

Increases Energy Efficiency

- Most processing power is tasked toward useful work of processing transactions

Increases Security

- An attack cannot revert a finalized block

CasperLabs has engineered the **first provably live and safe** implementation of CBC Casper.

Friendly for Developers to Build On

Built-In GraphQL Interface

Developers get instant and reliable insights about their application state. Confidently run decentralized applications, no hacks required.

Account Management for Humans

Users can be themselves with human readable account names, so it's easier to build and use blockchain apps. Parent-child account structure makes it easy to recover lost funds from any child account.

Familiar Languages, Tools & Workflows

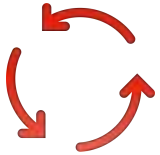
Written in Rust, allowing developers to write blockchain apps in any Web Assembly (WASM) compatible language. Over 95% of developers can write a blockchain app without training.

Future-Proof for Business

Predictable transaction fees

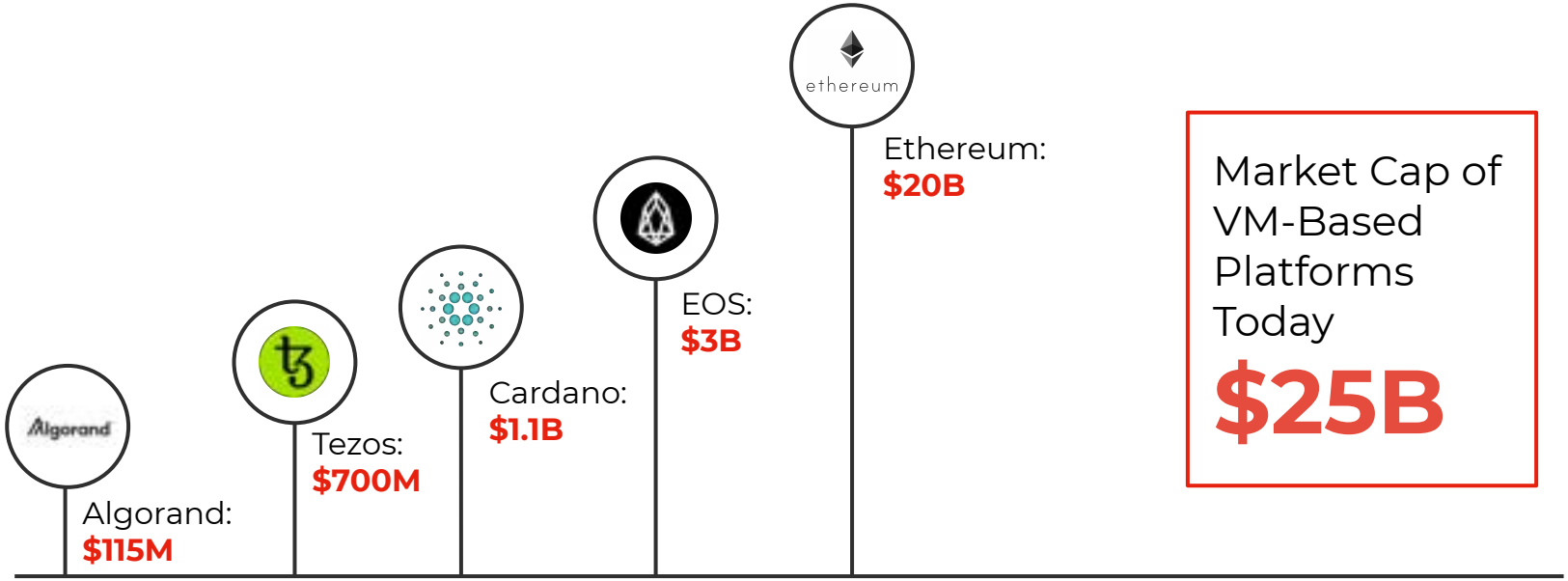
Businesses should not become victims of the network's success. Our economic model makes gas fees predictable so developers can build a business model -- and a business -- reliably.

Upgradeable infrastructure



Contracts can be upgraded easily with built-in versioning. Existing blocks remain immutable and all future blocks will exhibit the new contract behavior. Now, apps can evolve alongside business.

We Are Only At The Beginning



Market Cap of VM-Based Platforms Today

$25B

Adoption remains low:

3,000
dapps

Ethereum = 96% of all dapps

175,000
daily users

EOS = 71% of all DAUs

4,600
smart contracts

Ethereum = 83% of all smart contracts

Building Momentum Toward the Adoption Breakthrough

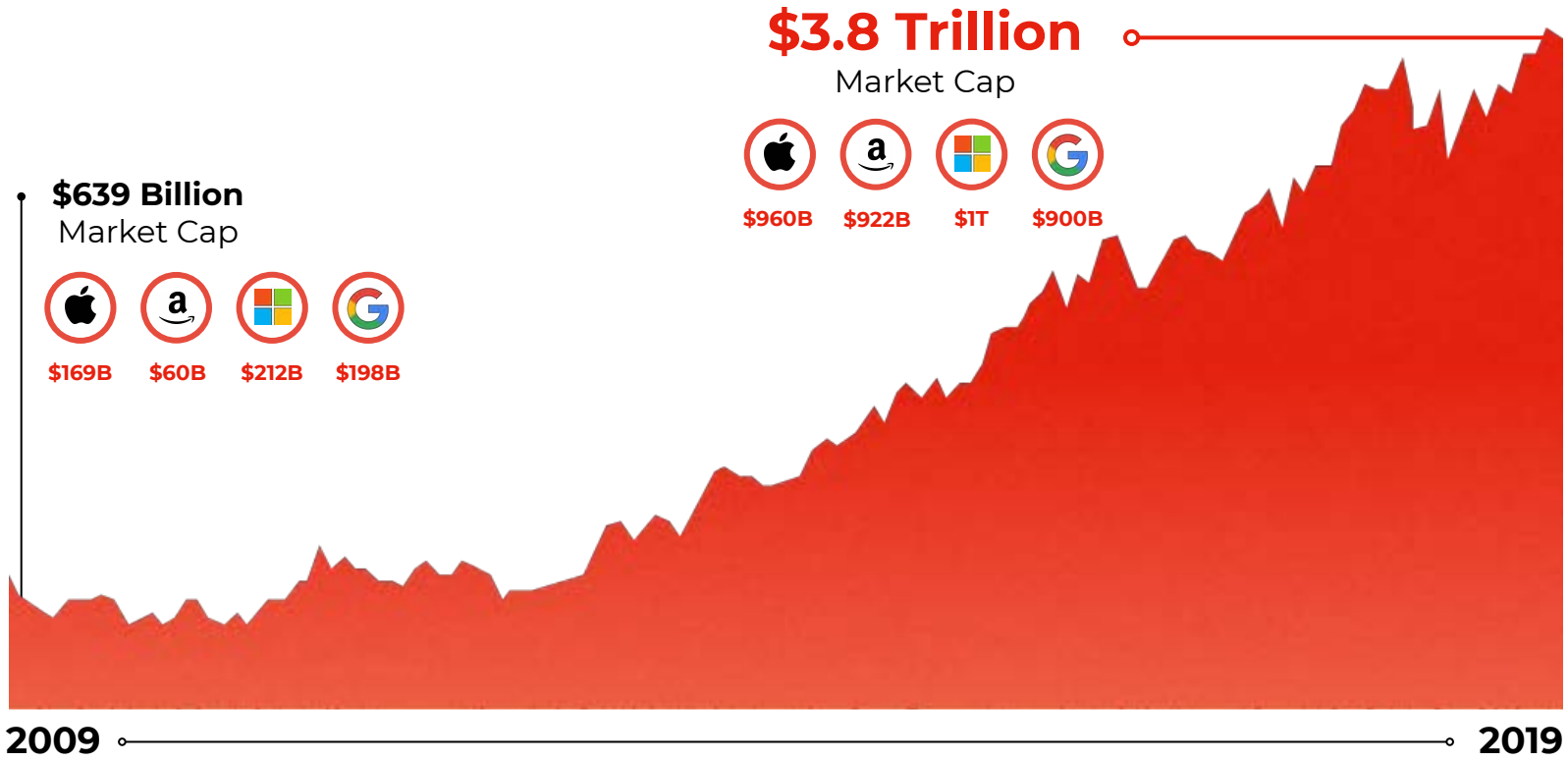
It took Amazon Web Services (AWS)
9 years to reach 1 Million Developer Accounts.

It took Apple App Store
5 years to reach 1 Million Apps.

It took Microsoft Office 365
6 years to reach 70 Million active users.

It took Google G Suite
11 years to reach 4 Million business accounts.

Platforms that Reach Scale are Massively Rewarded



CasperLabs Product Milestones

ACCOMPLISHMENTS TO DATE

Node 0.1
March 2019

- Casper Consensus
- Block Passing
- In-memory DAG

Node 0.2
April 2019

- Persistent global state
- Cost of execution
- Minting

Node 0.3
May 2019

- State storage
- gRPC Gossip

Node 0.4
May 2019

- GraphQL interface

Node 0.5
June 2019

Developer Network Preview

Node 0.6
August 2019

- CLarity - Block Explorer
- Payment codes
- Key management

Node 0.7
September 2019

- Deploy command arguments
- Faster voting
- Persistent deploy buffer

Consensus Protocol
September 2019

- Publish complete protocol spec with liveness and safety

Launch to Developers
October 2019

Genesis Validator Onboarding
November 2019

Test Network Launch
Q1-2020

Main Network Launch
Q2/Q3-2020

UPCOMING MILESTONES

Company Highlights

Oct 2018 - Nov 2019

Strong Team & Culture Growth

27 world-class engineers and platform business experts from **top tech companies**.

Google Microsoft amazon Dropbox

June 2019

Global Industry Partners

HDAC Technology
(Hyundai Digital Asset Company)

Agree to jointly research and develop consensus mechanisms and seamless integration between the HDAC and CasperLabs blockchains.

July 2019

Series A Equity Raise

$14.5M

Notable investors include:

Terren Peizer
Arrington XRP Capital
HashKey Capital
One Boat Capital
Consensus Capital

Why CasperLabs Wins

1. We are **founder-led**, guided by open source principles and radical transparency.

2. We have a diverse group of thinkers and builders with **proven experience** growing massive platforms.

3. We built the most robust blockchain network, scaling **opportunity without sacrifice**.

4. We are focused on rapid, **real-world adoption**.

5. We have the **future-proof foundation** to support enduring innovation and growth.

An Expert Team of Builders and Thinkers

MEDHA PARLIKAR
Chief Technology Officer
Product and engineering leader with decades of experience delivering production software at marquee companies such as Adobe, Omniture, Avalara, MP3.com and DivX.

MICHAEL BIRCH
Software Development Lead
Birch has written a working implementation of the Casper Protocol. He's also an accomplished functional programmer and mathematician. MS in physics from McMaster University.

ANDREAS FACKLER
Rust Developer
Fackler is an experienced Rust developer and has previously led teams at Google, MaidSafe and Xored's RCPTT. He has his PhD in Mathematics from Ludwig-Maximilians Universität.

DR. DANIEL KANE
Consensus Protocol Advisor
Dr. Kane has earned two Bachelor's Degrees from MIT and a Ph.D. from Harvard University. He is advising CasperLabs on blockchain cryptography and security.

MRINAL MANOHAR
Chief Executive Officer
Technology and finance leader previously with Microsoft, Bain, and Bain Capital. Ran the TMT sector of a $1B+ hedge fund. Early investor in Ethereum, Blockstack, etc.

CLIFF SARKIN
Chief Operating Officer
Entrepreneur and former VP of Biz Dev at DNA.Fund, a leading crypto fund. Led VideoSurf's $80M sale to Microsoft. J.D. from Harvard Law School.

YESSIN SCHIEGG
Chief Financial Officer
Seasoned accountant (PwC) and 10+ in private equity, typically as a CFO or similar. Advisory Board of the Ethereum Foundation, CFO of Status.im and Web3 Foundation.

SCOTT WALKER
Managing Director
Serial entrepreneur and angel investor. Founder/CEO of Atrinsic, a mobile game company he grew to $100M revenue. Early investor in BTC, ETH, and EOS.

Supported by a Circle of Luminaries and Innovators

ANIL ANTONY
Technology
& Social Entrepreneur

DENG CHAO
CEO
Hashkey Capital

STEVEN CHEN
Co-Founder/Managing Partner
Noris Capital

MERVYN CHNG
Co-Founder
MW Partners

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Founder
IOSG Capital

MATTI LIUKAS
CEO
Consensus Capital

RAN NEUNER
Partner - Onchain Capital
Reporter – CNBC Crypto
Trader

OMER OZDEN
Chairman
Rocktree Capital

WESLEY PRYOR
Founder
Liquid Solutions Malta

CRYSTAL ROSE
CEO
Sense.Chat

We Are Focused on Adoption

- **Attract builders** to build and adopt with lighthouse accelerators, partnerships, and programs.
- Make building **fast and simple** with a comprehensive platform of software, services, and support.
- Create customer and community **superfans** that scale our story.
- Partner with leading global enterprises to **upgrade entire industries**.

Bringing Builders to the Platform Through Global Accelerators

CV Labs, Plug and Play, TechStars
More than **300 startups** in the pipeline

A plan for global presence in 2020

- New York, NY
- San Francisco, CA
- Austin, TX
- Beijing, China
- Shanghai, China
- Vancouver, Canada
- Toronto, Canada
- Zug, Switzerland
- Jerusalem, Israel

CASPERLABS

Thank You.

Let’s start building.

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https://casperlabs.io

CONTACT US:
hello@casperlabs.io

t.me/casperlabs

@meetCasperLabs

Join us on WeChat

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